INCH-POUND

MIL-M-28787/193A
AMENDMENT 5
30 July 1993
SUPERSEDING
AMENDMENT 4
2 August 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, DIGITAL, KEY CODE SYT

This amendment forms a part of Military Specification MIL-M-28787/193A(EC), dated 3 August 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 8

*TABLE II, Test conditions E1-E8 and F1-F30, Test type column: Delete "X" and substitute "S".

TABLE II, Test conditions H1, H2, H7-H16, H21-H28, Min columns: Delete "-0.01" (four places) and substitute "0.0" (four places).

TABLE II, Test conditions H3-H6, H17-H20, Min columns: Delete "-0.01" (four places) and substitute "0.0" (four places).

PAGE 9

TABLE II, Test conditions H29, H30, Min columns: Delete "-0.01" (four places) and substitute "0.0" (four places).

PAGE 12

*TABLE II, Test conditions W4-W7, W19-W22, Max limit: Delete "15" (four places) and substitute "20" (four places).

PAGE 13

4.3: Delete in its entirety (including listing) and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

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PAGE 13

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, B, C, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, E, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, E, and F where applicable, followed by the remaining tests in the sequence."

PAGE 14

*4.5.1, line 5: Delete "D through V" and substitute "D, G through V".

*4.5.3, line 3: Delete "D through V" and substitute "D, G through V".

*4.5.3, line 5: Delete "D through Y" and substitute "D, G through Y".

6.3: Delete and substitute the following:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/193 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing 10001-2833769. Part number M28787/193-1 supersedes part numbers 2833769 and M28787/193."

PAGE 22

FIGURE 8: Delete and substitute new figure 8 as printed on page 5 of this amendment.

PAGE 34

FIGURE 12: Delete and substitute new figure 12 as printed on page 6 of this amendment.

PAGE 36

FIGURE 13: Delete and substitute new figure 13 as printed on page 7 of this amendment.

PAGES 38 AND 39

TABLE X, Test conditions H1 through H30, Min column: Delete "-0.01" (four places) and substitute "0.0" (four places).

PAGE 49

FIGURE 18: Delete and substitute new figure 18 as printed on page 8 of this amendment.

PAGE 53

FIGURE 19, ITEM P01-2: Delete "2.0V ± 1%" and substitute "3.5V ± 1%".

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PAGE 56

TABLE XVI: Delete and substitute new table XVI as printed on page 9 of this amendment.

PAGE 67

FIGURE 25: Delete and substitute new figure 25 as printed on page 10 of this amendment.

PAGE 75

FIGURE 28, NOTE 4: Delete "t_{WP} = 40 ± 2 ns" and substitute "t_{WP} = 40 ± 2 ns; 50 ± 2 ns for 150 pF load at 60°C".

FIGURE 28, NOTE 11: Delete "2.2" and substitute "2.5".

FIGURE 28, NOTE 12: Delete "0.6 ± 0.2V" and substitute "0.45 ± 0.05V".

PAGE 78

FIGURE 29: Delete and substitute new figure 29 as printed on page 11 of this amendment.

PAGE 82

FIGURE 32, NOTE 4: Delete "t_{WP} = 40 ± 2 ns" and substitute "t_{WP} = 40 ± 2 ns; 50 ± 2 ns for 150 pF load at 60°C".

FIGURE 32, NOTE 11: Delete "2.2" and substitute "2.5".

FIGURE 32, NOTE 12: Delete "0.6 ± 0.2V" and substitute "0.45 ± 0.05V".

PAGE 86

FIGURE 33: Delete and substitute new figure 33 as printed on page 12 of this amendment.

PAGE 92

FIGURE 36, NOTE 11: Delete "2.2" and substitute "2.5".

FIGURE 36, NOTE 12: Delete "0.6 ± 0.2V" and substitute "0.45 ± 0.05V".

PAGE 96

FIGURE 37: Delete and substitute new figure 37 as printed on page 13 of this amendment.

PAGE 101

FIGURE 40, NOTE 4: Delete "t_{WP} = 40 ± 2 ns" and substitute "t_{WP} = 40 ± 2 ns; 50 ± 2 ns for 150 pF load at 60°C".

FIGURE 40, NOTE 13: Delete "2.2" and substitute "2.5".

FIGURE 40, NOTE 14: Delete "0.6 ± 0.2V" and substitute "0.45 ± 0.05V".

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PAGES 105 and 106

*TABLE XXIII, Test conditions W4-W7, W19-W22, Max limit: Delete "15" and substitute "20".

PAGE 109

*FIGURE 43, ITEM C01 box: Delete "C01" and substitute "C06".

*FIGURE 43, NOTE 3, first line: Delete "(C01)" and substitute "(C06)".

FIGURE 43, NOTE 3: Delete "120 Hz" and substitute "1 kHz".

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

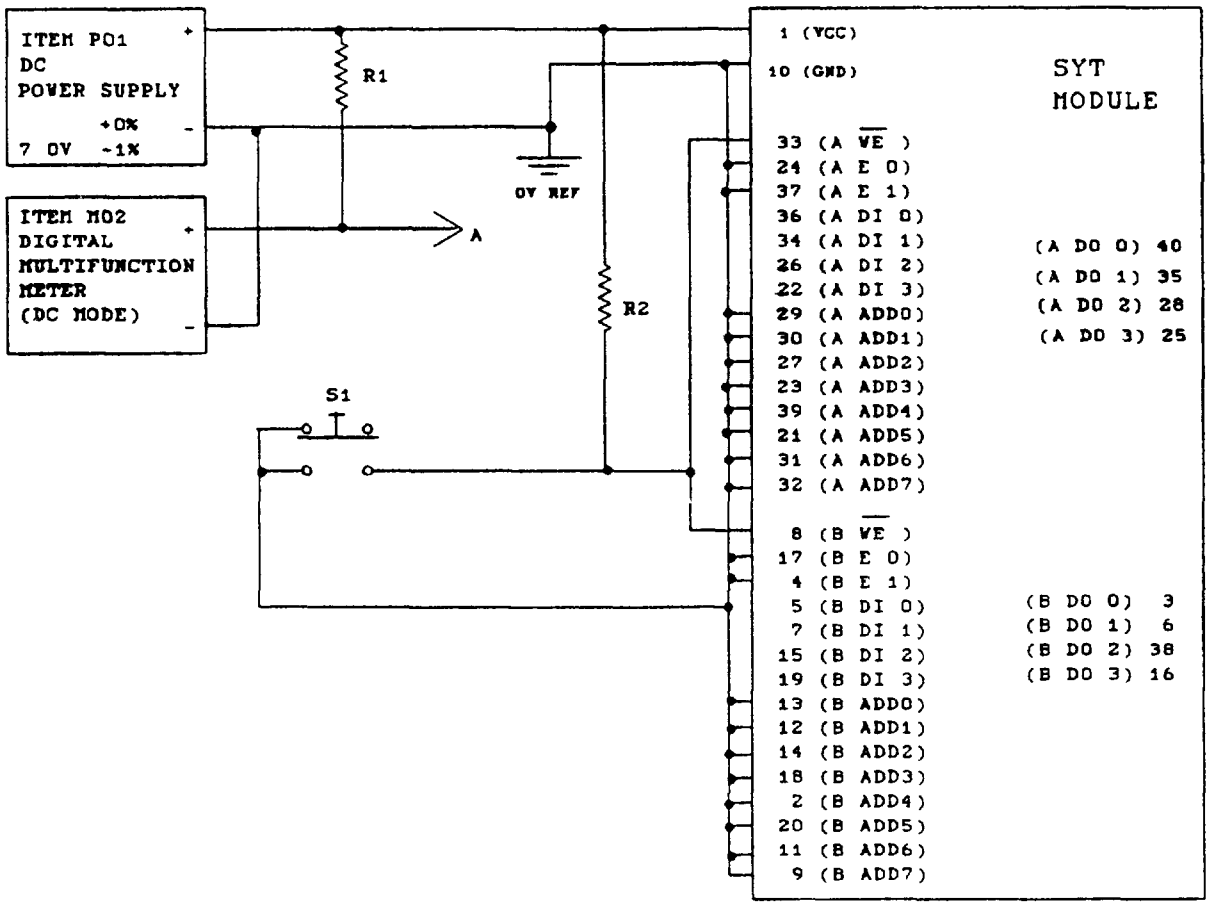
Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

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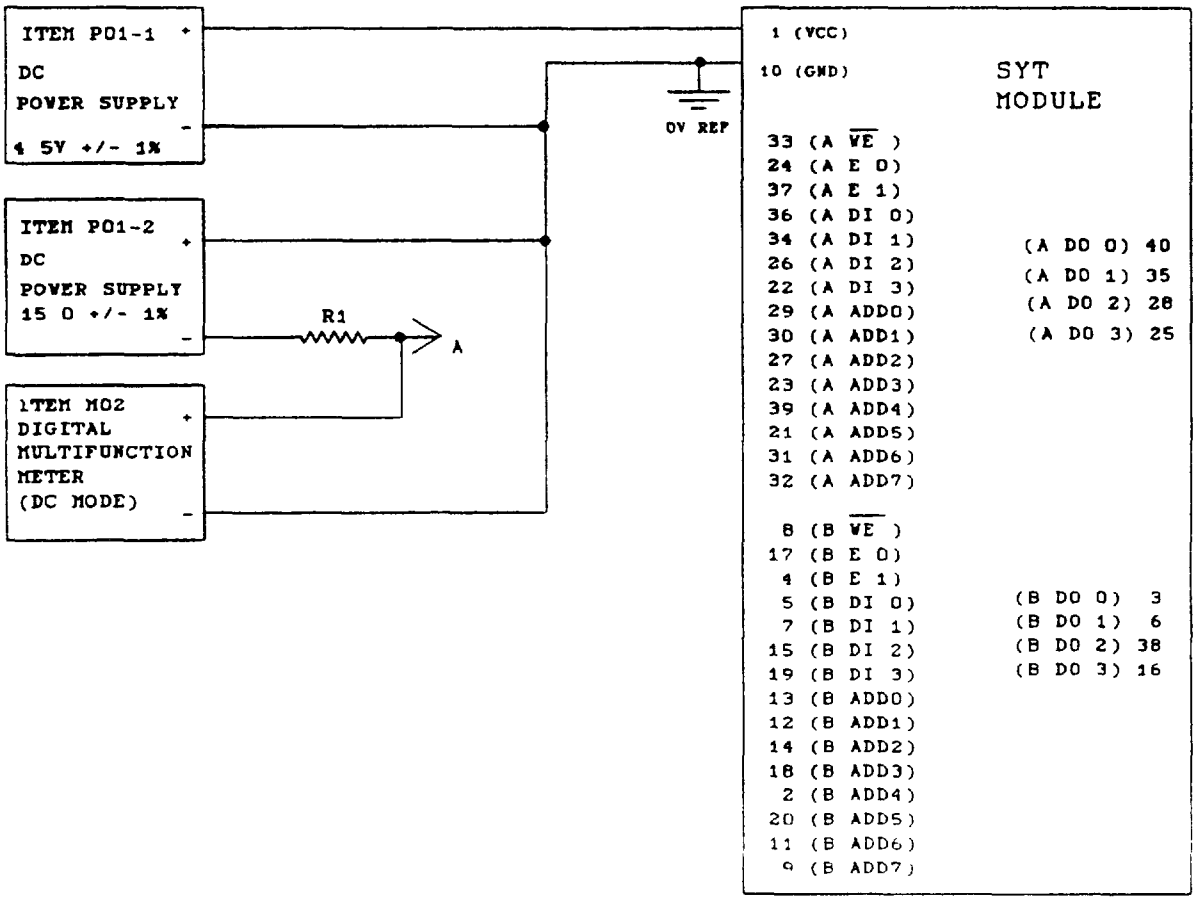
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NOTES:
1. R1 = 324Ω ± 1%, 1/8 W.
2. R2 = 1 kΩ ± 5%, 1/4 W.
3. S1 = push button.
4. Measurements shall be recorded after point A has been connected to the appropriate pin for a minimum of 5 seconds.
5. S1 shall be pushed and released once before starting the test.

FIGURE 8. Vout (low-loaded) test circuit for table IV.

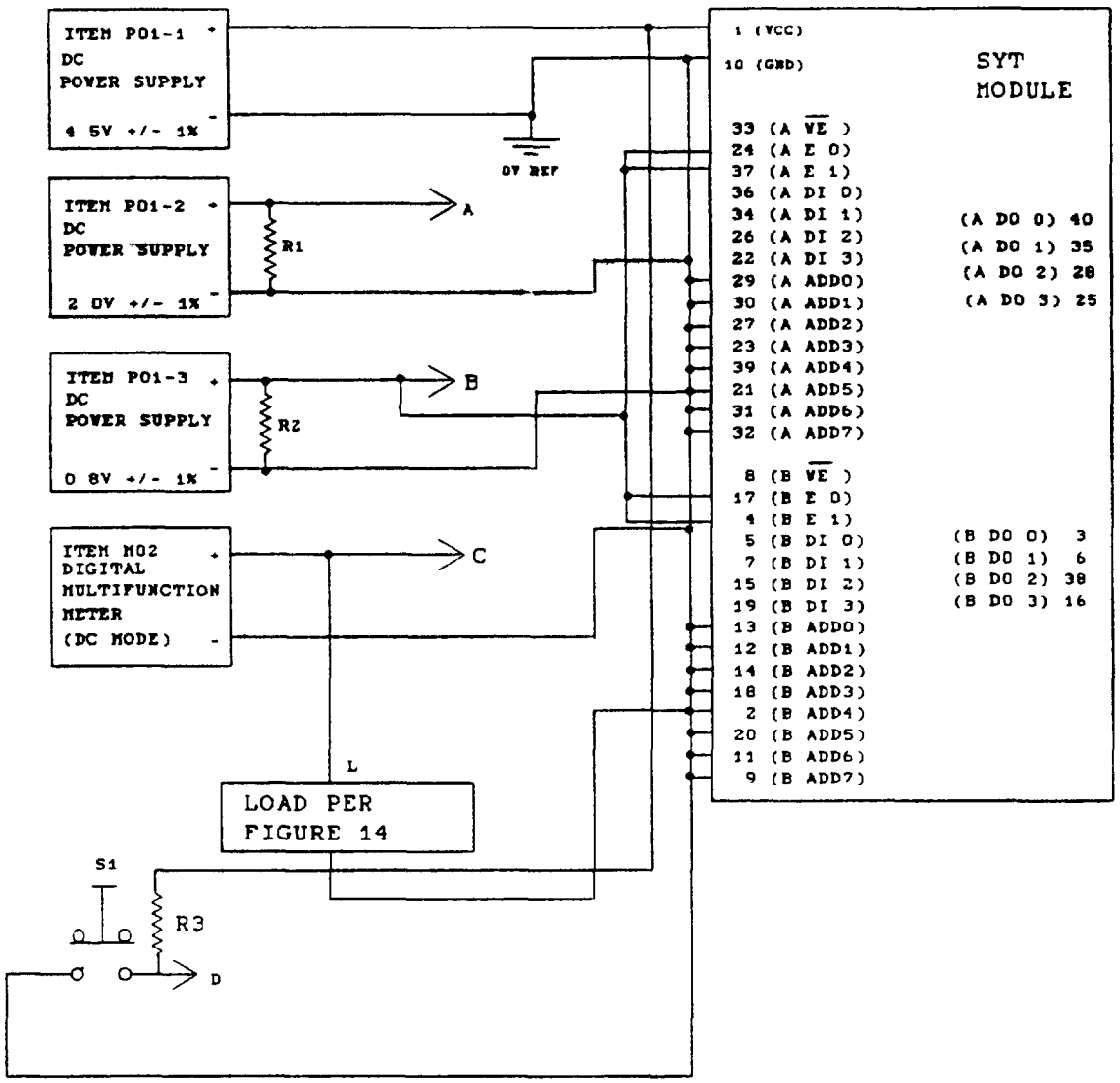
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NOTE: R1 = 1.15 kΩ ± 1%, 1/4 W.

FIGURE 12. Input clamp diode voltage test circuit for table VIII.

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NOTES:
1. R1 = R2 = 51Ω ± 5%, 1 W.
2. R3 = 1 kΩ ± 5%, 1/4 watt.
3. S1 = push button.
4. S1 shall be pushed and released before each test condition.

FIGURE 13. Static output voltage (low-level and high-level) test circuit for table IX.

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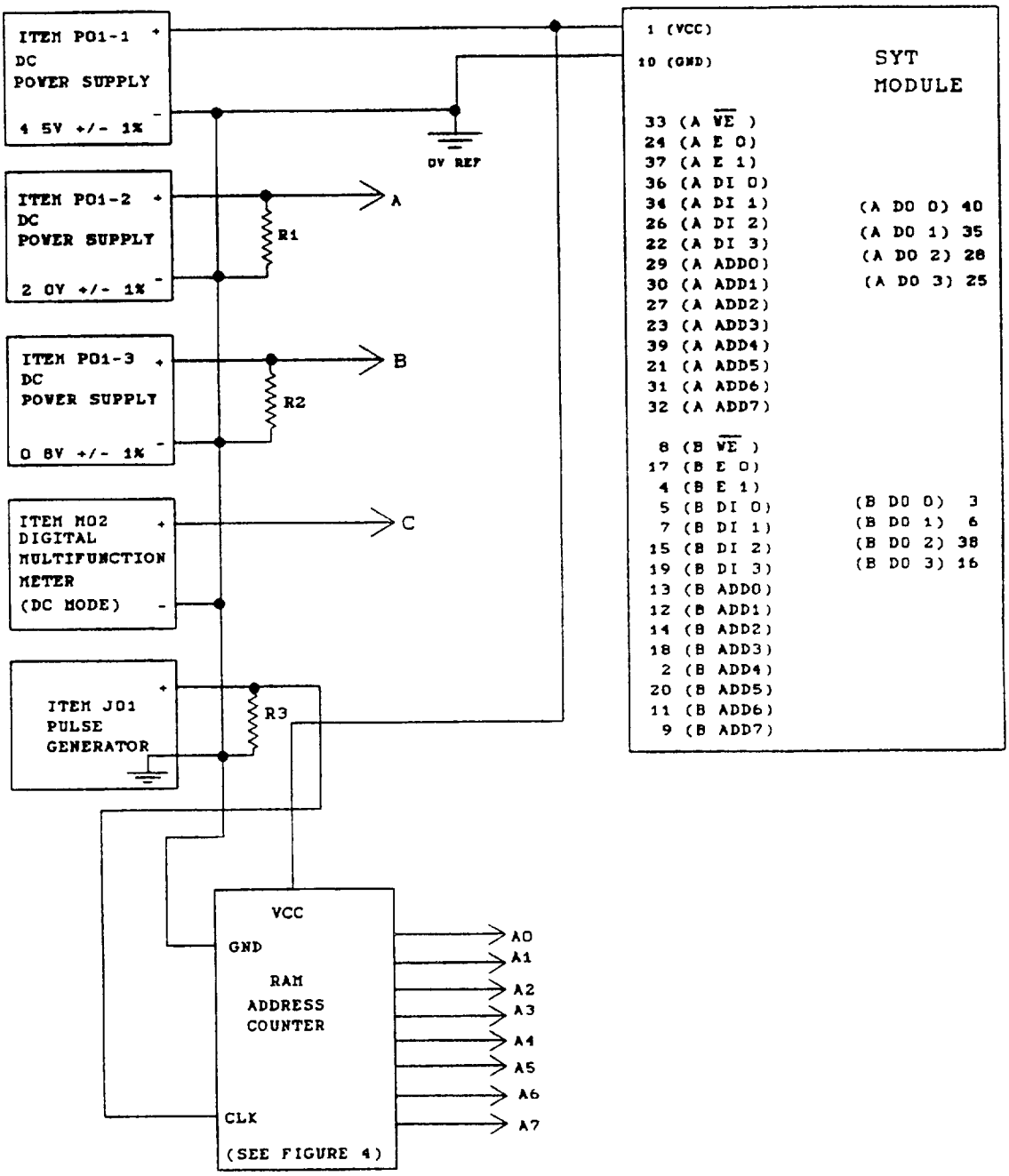


FIGURE 18. Address threshold (low-level and high-level) test circuit for tables XIII and XIV.

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TABLE XVI. Enable recovery time (low-state to high-impedance state tERL and high-state to high-impedance state tERH) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	AD00 Enable recovery time (tERL)	24,37	29	40	3	30	ns	21,22
P2	AD01			35				
P3	AD02			28				
P4	AD03	24,37	29	25				
P5	BD00	4,17	13	3				
P6	BD01			6				
P7	BD02			38				
P8	BD03 Enable recovery time (tERL)	4,17	13	16				
P9	AD00 Enable recovery time (tERH)	24,37	29	40				
P10	AD01			35				
P11	AD02			28				
P12	AD03	24,37	29	25				
P13	BD00	4,17	13	3				
P14	BD01			6				
P15	BD02			38				
P16	BD03 Enable recovery time (tERH)	4,17	13	16	3	30	ns	21,22

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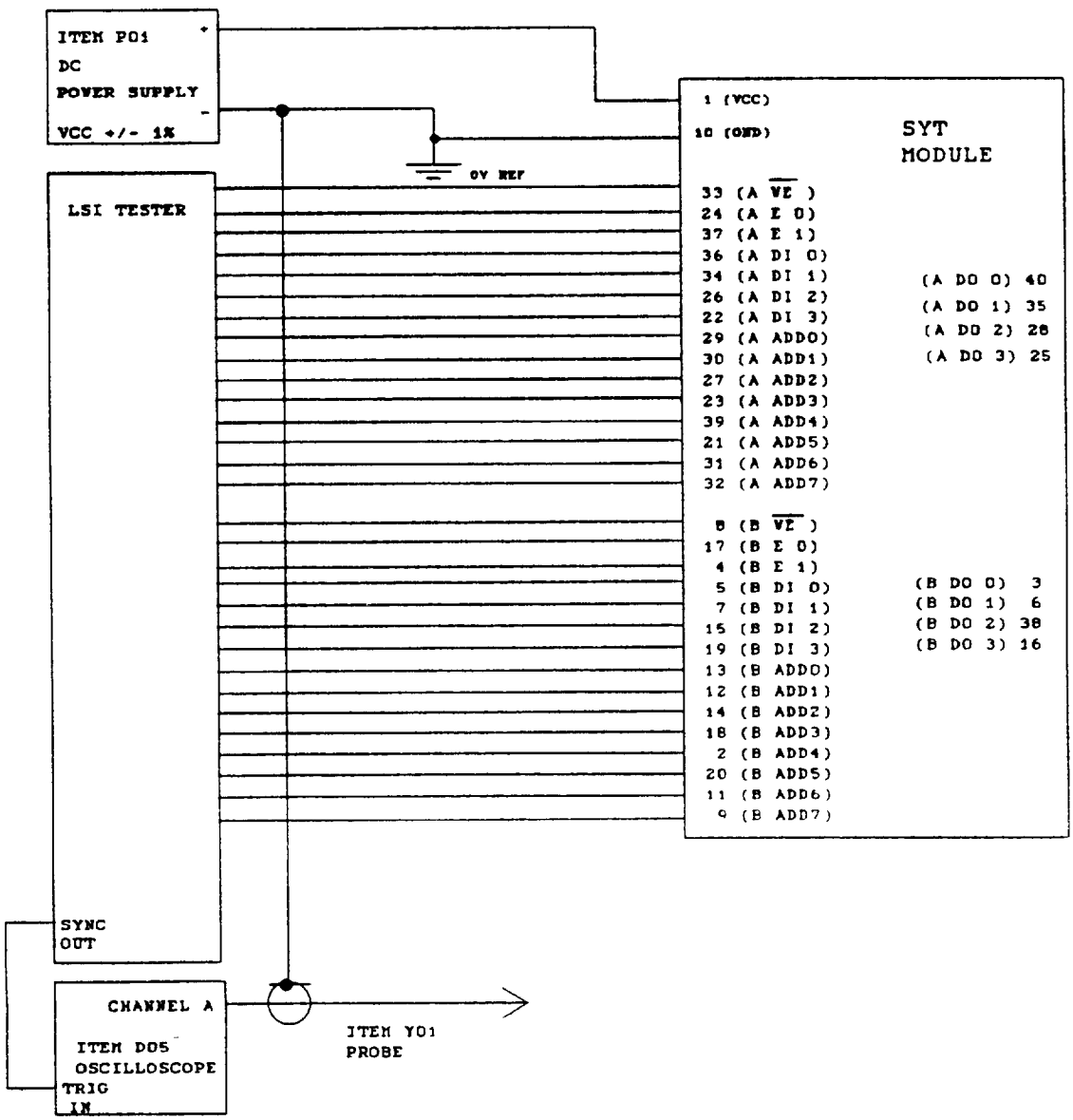


FIGURE 25. Walking-one-walking-zero test pattern circuit for table XVIII.

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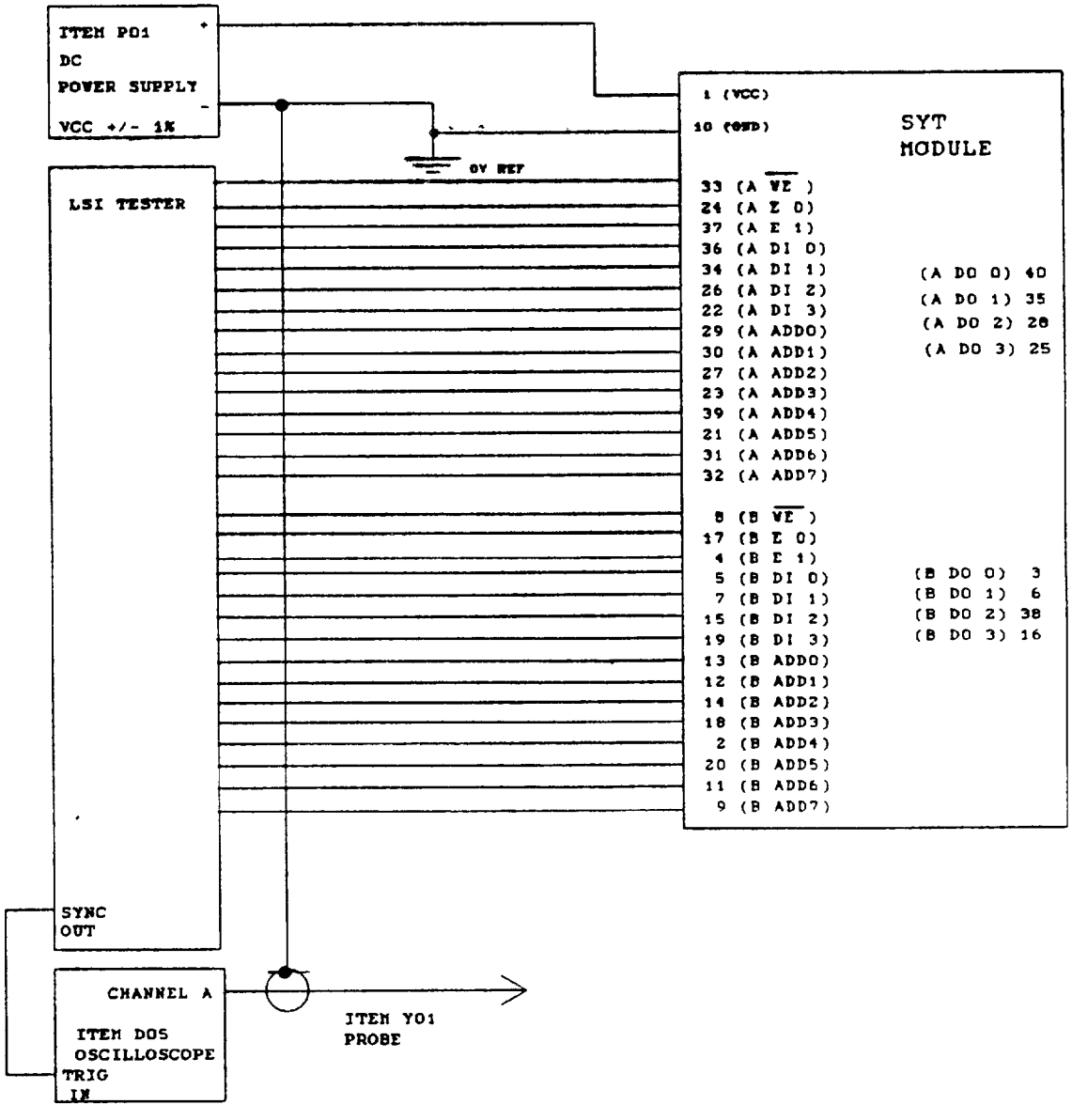


FIGURE 29. Galloping-one-galloping-zero test pattern circuit for table XIX.

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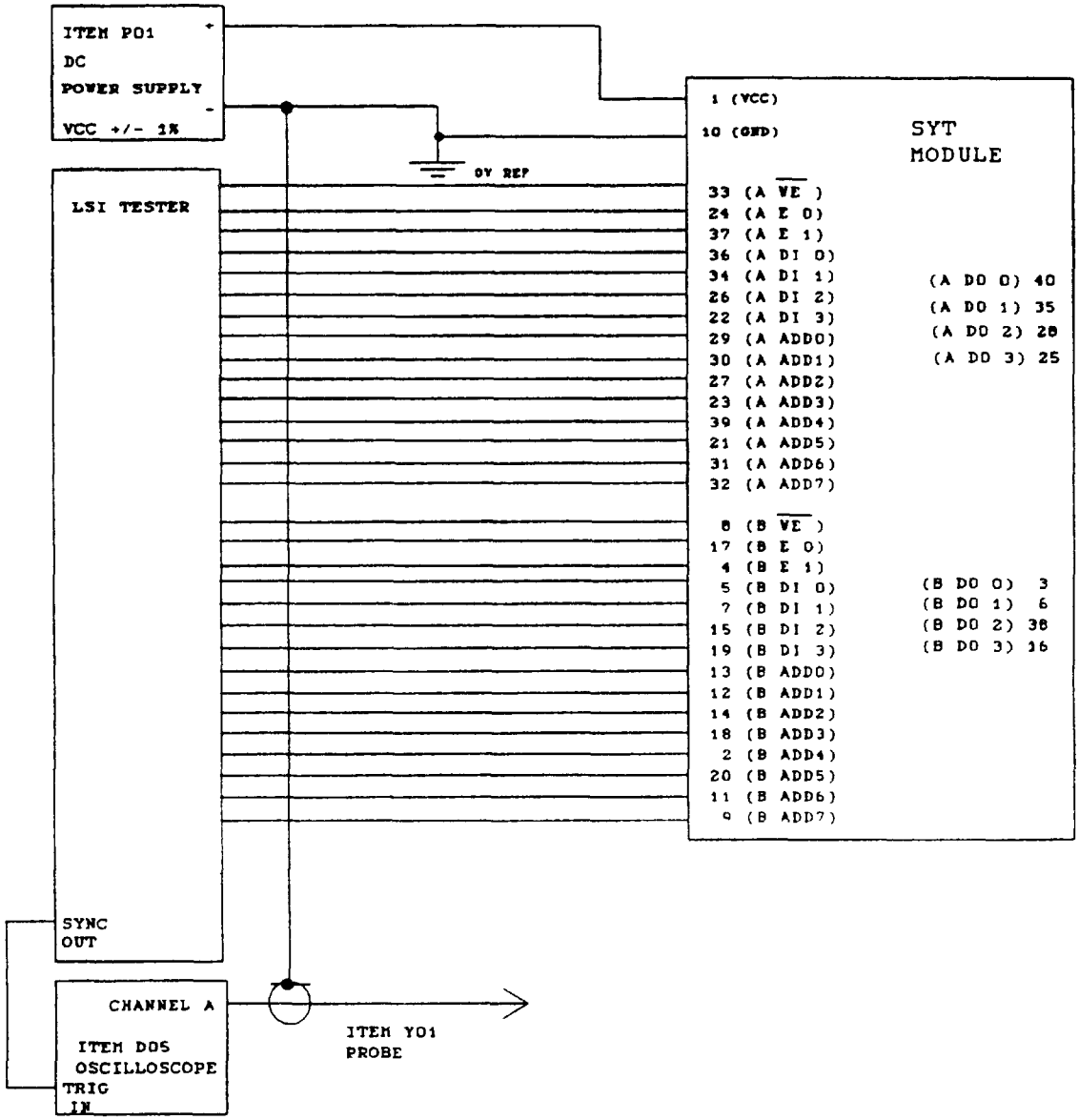


FIGURE 33. Inter-address write-recovery-one-write-recovery-zero test pattern circuit for table XX.

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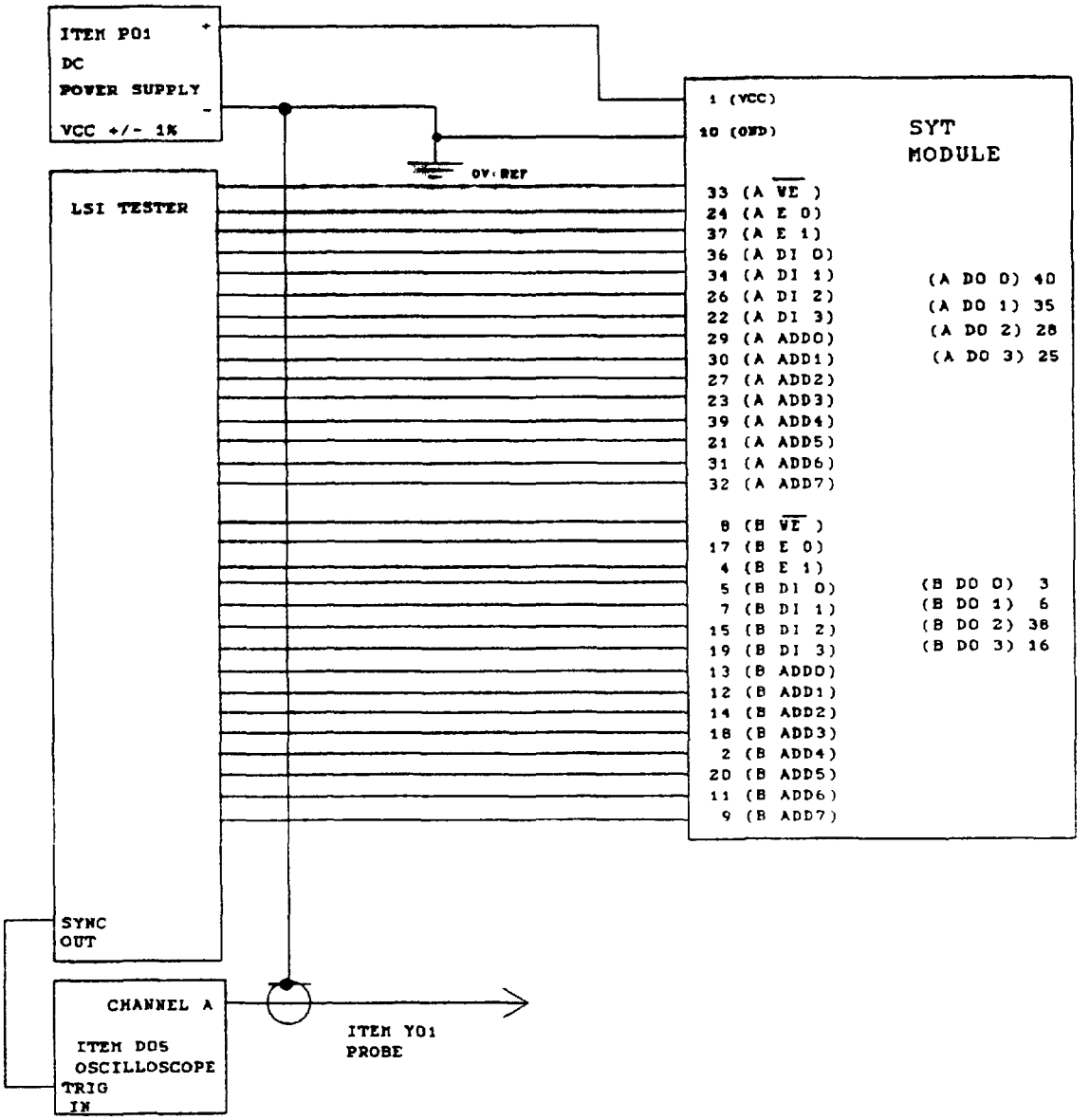


FIGURE 37. Marching intra-address write-recovery-one-write-recovery-zero and enable/disable test pattern for table XXI.

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